

Binomial random variable

$X \sim \text{Binomial}(n, p)$

$$E(X) = np$$

$$\text{Var}(X) = np(1 - p)$$

Exponential random variable

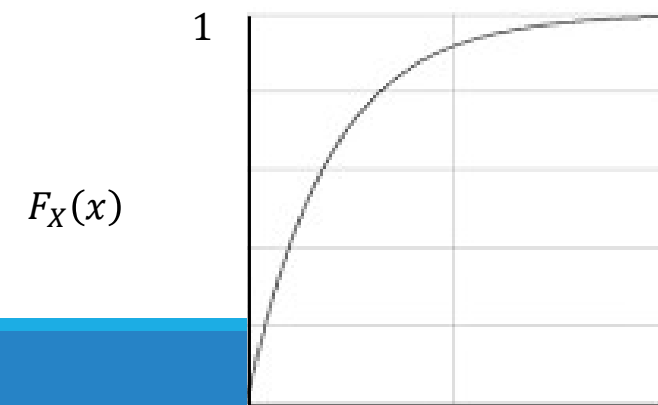
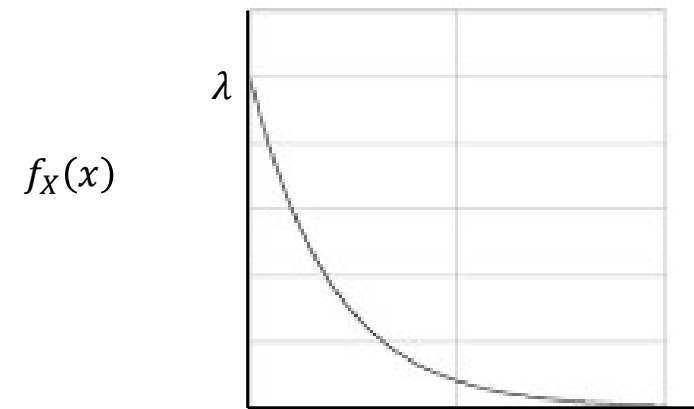
$$\text{PDF: } f_X(x) = \begin{cases} \lambda e^{-\lambda} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{CDF: } F_X(x) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$E[X] = \mu = 1/\lambda$$

$$\text{Var}(X) = 1/\lambda^2$$

$$\text{Skewness of } X = 2$$



Statistics of a random sample

$(X_1, X_2, \dots, X_n)$: a random sample

Sample mean:

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

Sample variance:

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$$

Sample standard deviation, S : positive square root of S^2

Linear Combinations of Measurements

$(X_1, X_2, \dots, X_n)$: a possibly correlated sample with mean μ_i and variance σ_i^2

$Y = \sum_{i=1}^n c_i X_i$: function of $(X_1, X_2, \dots, X_n)$, where c_i 's are constants

$E(Y)$ and $\text{Var}(Y)$: ?

$$E(Y) = E\left(\sum_{i=1}^n c_i X_i\right) = \sum_{i=1}^n c_i \mu_i$$

$$\text{Var}(Y) = \text{Var}\left(\sum_{i=1}^n c_i X_i\right) = \sum_{i=1}^n c_i^2 \sigma_i^2 + 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n c_i c_j \text{Cov}(X_i, X_j)$$

Mean Square Error of an Estimator

$$MSE(\hat{\Theta}) = E[(\hat{\Theta} - \theta)^2] = \underbrace{E[(\hat{\Theta} - E[\hat{\Theta}])^2]}_{\text{Variance of } \hat{\Theta}} + \underbrace{(E[\hat{\Theta}] - \theta)^2}_{(\text{Bias})^2} = (\sigma_{\hat{\theta}})^2 + (\mu_{\hat{\theta}} - \theta)^2$$

$\sigma_{\hat{\theta}} = \sqrt{V(\hat{\theta})}$ is called the standard error of an estimator $\hat{\theta}$.

Methods of Maximum Likelihood

Suppose that X is a random variable with probability distribution $f(x; \theta)$, where θ is a single unknown parameter. Let $x_1, x_2, \dots, x_n$ be the observed values in a random sample of size n . Then the **likelihood function** of the sample is

$$L(\theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdot \dots \cdot f(x_n; \theta) \quad (7-5)$$

Note that the likelihood function is now a function of only the unknown parameter θ . The **maximum likelihood estimator** of θ is the value of θ that maximizes the likelihood function $L(\theta)$.

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy$$

Central Limit Theorem (CLT)

$X_1, \dots, X_n$ be independent identically distributed (iid) rvs with $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$. Then:

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} \rightarrow Z_n \sim N(0,1) \quad \text{as } n \rightarrow \infty,$$

OR

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{\sum_{i=1}^n X_i/n - \mu}{\sigma/\sqrt{n}} \rightarrow Z_n \sim N(0,1) \quad \text{as } n \rightarrow \infty.$$

Parameter of interest: true mean, μ

- $X_1, \dots, X_n$: A random sample with $E[X] = \mu$
- We are looking for a random variable $Y=h(X_1, \dots, X_n; \mu)$ whose distribution can be specified.

	X is normal		X is not normal		
	$n < 40$	$n \geq 40$	$n < 30$	$30 \leq n < 40$	$n \geq 40$
σ : known	$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim Z$	$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim Z$	?	$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim Z$	$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim Z$
σ : unknown	$\frac{\bar{x} - \mu}{s/\sqrt{n}} \sim T_{n-1}$	$\frac{\bar{x} - \mu}{s/\sqrt{n}} \sim Z$	?	? $\frac{\bar{x} - \mu}{s/\sqrt{n}} \sim T_{n-1}$?	$\frac{\bar{x} - \mu}{s/\sqrt{n}} \sim Z$

CIs around the true mean, μ

- $X_1, \dots, X_n$: A random sample with $E[X] = \mu$
- We are looking for a random variable $Y=h(X_1,\dots,X_n; \mu)$ whose distribution can be specified.

	X is normal		X is not normal		
	n<40	n≥40	n<30	30≤n<40	n≥40
σ : known	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	?	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
σ : unknown	$\bar{x} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$	?	? $\bar{x} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$?	$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$

A Confidence Interval for the Difference Between Two Means

$X_1, \dots, X_m$: a random sample from a population with μ_X & σ_X^2
 $Y_1, \dots, Y_n$: a random sample from a population with μ_Y & σ_Y^2

} X & Y samples are independent of one another.

Samples: normal
 σ_X and σ_Y : known

100(1 - α)% CI
for $\mu_X - \mu_Y$ is:

$$\bar{X} - \bar{Y} \pm z_{\alpha/2} \sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}.$$

Samples: any distribution
 σ_X and σ_Y : unknown
 n_X and n_Y : large

100(1 - α)% CI
for $\mu_X - \mu_Y$ is:

$$\bar{X} - \bar{Y} \pm z_{\alpha/2} \sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}$$

$$Z = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}}$$

$$Z = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}}$$

A Confidence Interval for the Difference Between Two Means

$X_1, \dots, X_{n_X}$: a random sample from a population with μ_X & σ_X^2
 $Y_1, \dots, Y_{n_Y}$: a random sample from a population with μ_Y & σ_Y^2

} X & Y samples
are independent
of one another.

$$T_v = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}}$$

$$v = \frac{\left(\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}\right)^2}{\frac{\left(\frac{s_X^2}{n_X}\right)^2}{n_X - 1} + \frac{\left(\frac{s_Y^2}{n_Y}\right)^2}{n_Y - 1}}$$

Samples: normal
 σ_X and σ_Y : unknown
 n_X and n_Y : small

100(1 - α)% CI
for $\mu_X - \mu_Y$ is:

$$\bar{X} - \bar{Y} \pm t_{v, \alpha/2} \sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}$$

If $n_X = n_Y = n$, then:

$$v = (n - 1) \left(1 + \frac{2s_X^2 s_Y^2}{s_X^4 + s_Y^4} \right)$$

A Confidence Interval for the Difference Between Two Means

$X_1, \dots, X_n$: a random sample from a population with μ_X
 $Y_1, \dots, Y_n$: a random sample from a population with μ_Y
 $D_1 = X_1 - Y_1, \dots, D_n = X_n - Y_n$: a random sample from a population with μ_D

} X & Y samples are positively correlated

$$T_{n-1} = \frac{\bar{D} - \mu_D}{\frac{s_D}{\sqrt{n}}}$$

Samples: normal

σ_X and σ_Y : unknown

n_X and n_Y : small

100(1 - α)% CI for $\mu_X - \mu_Y$ is:

$$\bar{D} \pm t_{n-1, \alpha/2} \frac{s_D}{\sqrt{n}}$$

Parameter of interest: true population proportion p

X : Number of successes in n trials, each trial is i.i.d. with constant $p = P\{\text{success}\}$

We are looking for a random variable, $Y = h(X; \theta)$ whose distribution can be specified.

$$\hat{p} = \frac{X}{n}$$

$n\hat{p} \geq 10 \text{ and } n(1 - \hat{p}) \geq 10$
$Z = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}}$

Traditional

CIs around the true population proportion p

X : Number of successes in n trials, each trial is i.i.d. with constant $p = P\{\text{success}\}$

We are looking for a random variable, $Y = h(X; \theta)$ whose distribution can be specified.

$$\hat{p} = \frac{X}{n}$$

$$\tilde{p} = \frac{X + 2}{n + 4}$$

$$\tilde{n} = n + 4$$

$$n\hat{p} \geq 10 \text{ and } n(1 - \hat{p}) \geq 10$$

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

No condition on n or p

$$\tilde{p} \pm z_{\alpha/2} \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{\tilde{n}}}$$

100(1- α) Confidence Interval around $p_X - p_Y$ New Approach

100(1 - α)% CI for the difference $p_X - p_Y$ is

$$\tilde{p}_X - \tilde{p}_Y \pm z_{\alpha/2} \sqrt{\frac{\tilde{p}_X(1 - \tilde{p}_X)}{\tilde{n}_X} + \frac{\tilde{p}_Y(1 - \tilde{p}_Y)}{\tilde{n}_Y}}$$

If the lower limit of the confidence interval is less than -1, replace it with -1.

If the upper limit of the confidence interval is greater than 1, replace it with 1.

$$\tilde{P}_X = \frac{X+1}{\tilde{n}_X}, \tilde{n}_X = n_X + 2, \tilde{P}_Y = \frac{Y+1}{\tilde{n}_Y}, \tilde{n}_Y = n_Y + 2$$

Inferences about the true variance of a random variable X

$X_1, \dots, X_n$: A random sample

We are looking for a test statistics, $Y=h(X_1, \dots, X_n; \theta)$ such that its probability distribution does not depend on θ or any other unknown parameters.

$X \sim \text{Normal}$		X is not normal	
$n < 40$	$n \geq 40$	$n < 40$	$n \geq 40$
$\chi^2_{n-1} = \frac{(n-1)S^2}{\sigma^2}$	$\chi^2_{n-1} = \frac{(n-1)S^2}{\sigma^2}$	—	—

Prediction interval (PI) for a single future value

$X_1, \dots, X_n$: a normal random sample from a population with μ_X & σ_X^2

$$T_{n-1} = \frac{\bar{X} - X_{n+1}}{S \sqrt{1 + 1/n}}$$

Samples: normal

μ_X & σ_X : *unknown*

100(1 - α)% PI for X_{n+1} is:

$$\left(\bar{x} \pm t_{n-1, \alpha/2} S \sqrt{1 + \frac{1}{n}} \right).$$

Tolerance Intervals

Let $X_1, \dots, X_n$ be a sample from a *normal* population.

A tolerance interval for containing $100(1 - \gamma)\%$ of the population with confidence $100(1 - \alpha)\%$ is:

$$\bar{X} \pm k_{n,\alpha,\gamma} S$$



Table A.4

Test statistics and their distributions (true mean, μ)

- $X_1, \dots, X_n$: A random sample with $E[X] = \mu$
- We are looking for a random variable $Y = h(X_1, \dots, X_n; \mu)$ whose distribution can be specified.

	X is normal		X is not normal		
	$n < 40$	$n \geq 40$	$n < 30$	$30 \leq n < 40$	$n \geq 40$
σ : known	$\frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \sim Z$	$\frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \sim Z$	?	$\frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \sim Z$	$\frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \sim Z$
σ : unknown	$\frac{\bar{x} - \mu}{s / \sqrt{n}} \sim T_{n-1}$	$\frac{\bar{x} - \mu}{s / \sqrt{n}} \sim Z$	?	? $\frac{\bar{x} - \mu}{s / \sqrt{n}} \sim T_{n-1}$?	$\frac{\bar{x} - \mu}{s / \sqrt{n}} \sim Z$

Test statistics and their distributions ($\mu_X - \mu_Y$)

$$\sigma = \sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}$$

$$s = \sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}$$

- $\{X_1, \dots, X_n\}, \{Y_1, \dots, Y_m\}$: Two random samples independent of each other with $E[X] = \mu_X$ and $E[Y] = \mu_Y$ with $\text{Var}(X) = \sigma_X$ and $\text{Var}(Y) = \sigma_Y$
- We are looking for a random variable $U = h(X_1, \dots, X_n; Y_1, \dots, Y_m; \mu_X, \mu_Y)$ whose distribution can be specified.

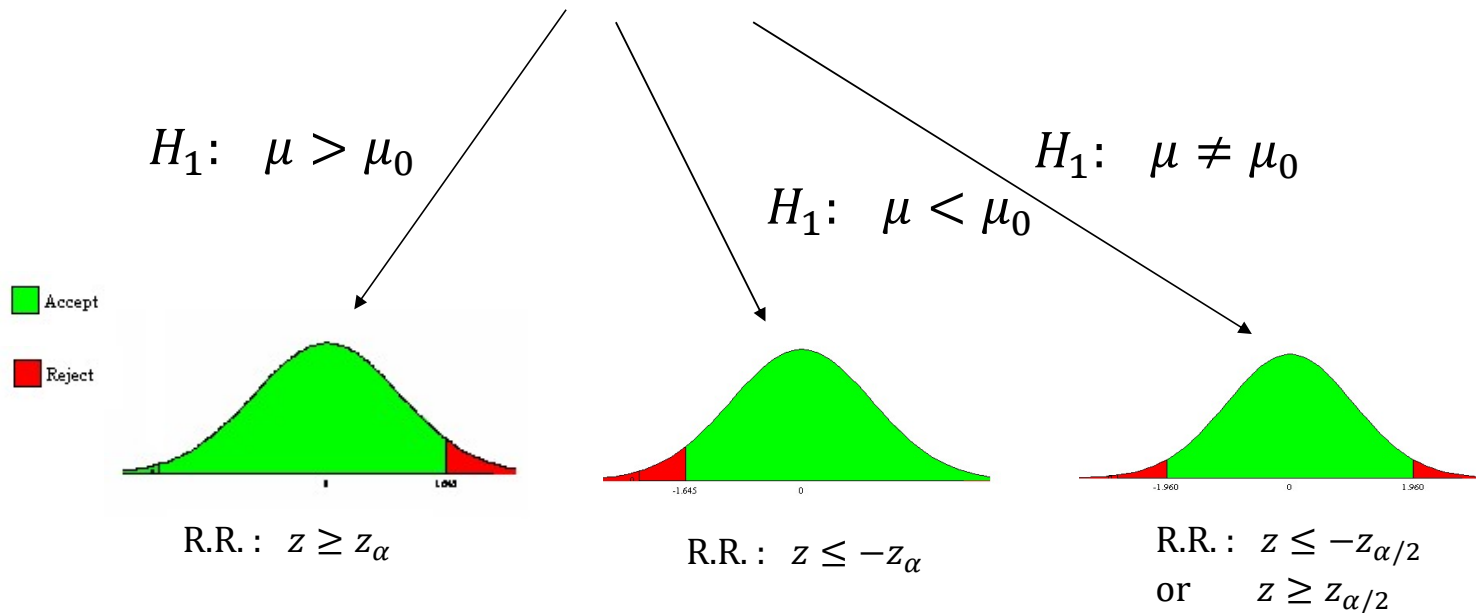
	X is normal		X is not normal		
	n<40	n≥40	n<30	30≤n<40	n≥40
σ: known	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{\sigma} \sim Z$	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{\sigma} \sim Z$	?	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{\sigma} \sim Z$	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{\sigma} \sim Z$
σ: not known	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{s} \sim T_{n-1}$	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{s} \sim Z$	?	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{s} \sim T_{n-1}$?	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{s} \sim Z$

Tests about a population mean of a normal r.v. with known variance

$$H_0: \mu = \mu_0$$

$$\text{Test statistics: } Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

Test statistics is given under the assumption that H_0 is true.

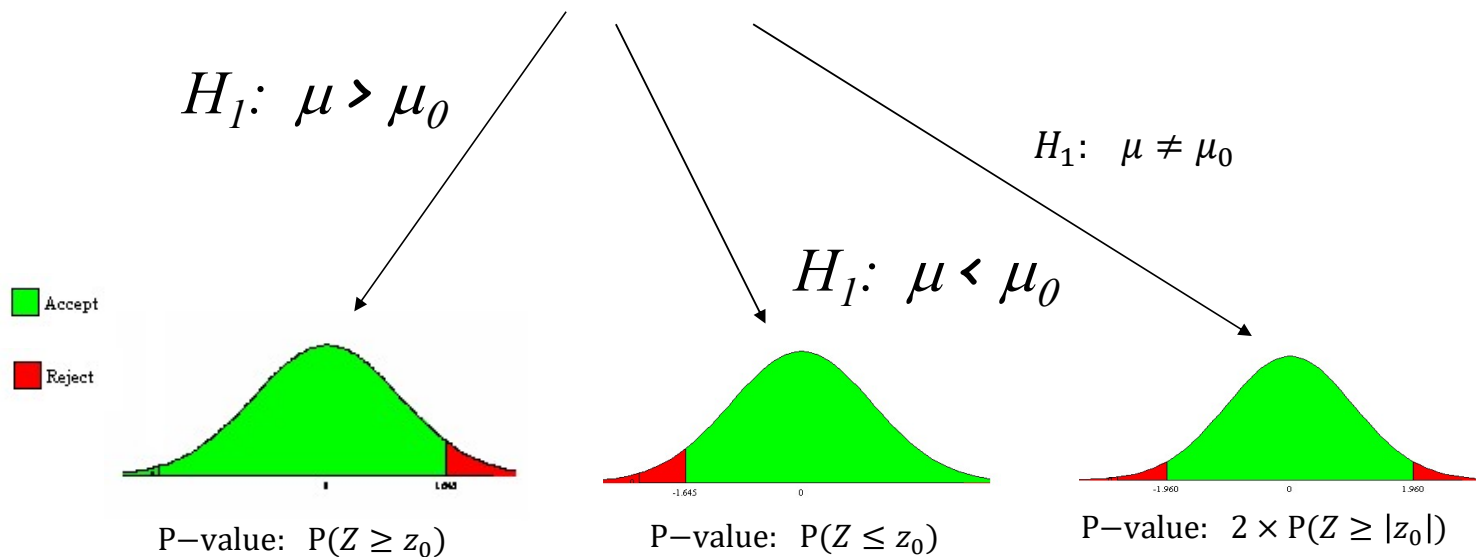


Tests about a population mean of a normal r.v. with known variance (Significance Testing)

$$H_0: \mu = \mu_0$$

$$\text{Test statistics: } z_0 = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Test statistics is given
under the assumption
that H_0 is true.

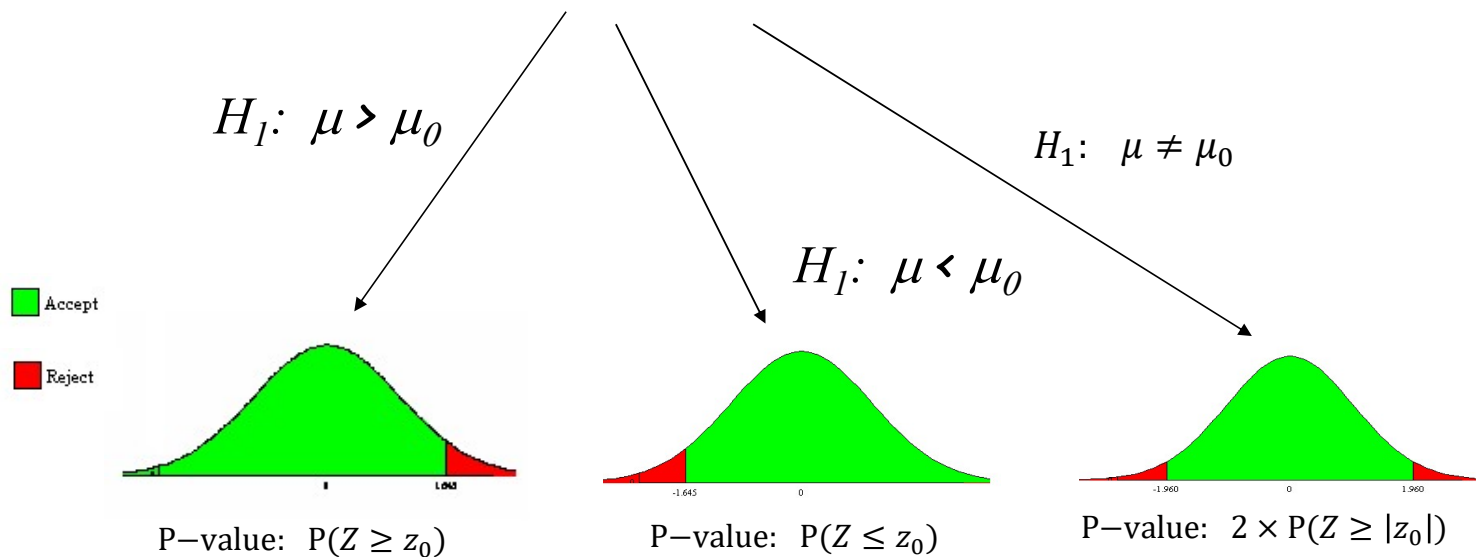


Tests about a population mean of a r.v. with large sample size

$$H_0: \mu = \mu_0$$

$$\text{Test statistics: } z_0 = \frac{\bar{x} - \mu_0}{S/\sqrt{n}}$$

Test statistics is given under the assumption that H_0 is true.

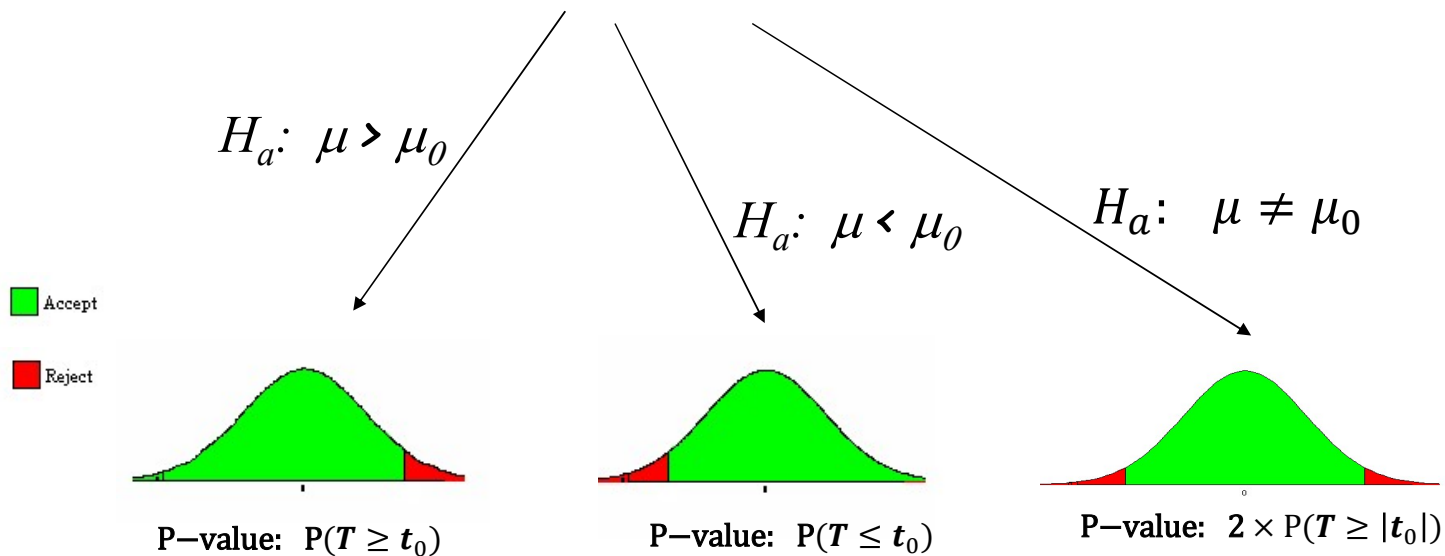


Tests about a population mean of a normal r.v. with unknown variance and small sample size

$$H_0: \mu = \mu_0$$

$$\text{Test statistics: } t_0 = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$$

Test statistics is given under the assumption that H_0 is true.

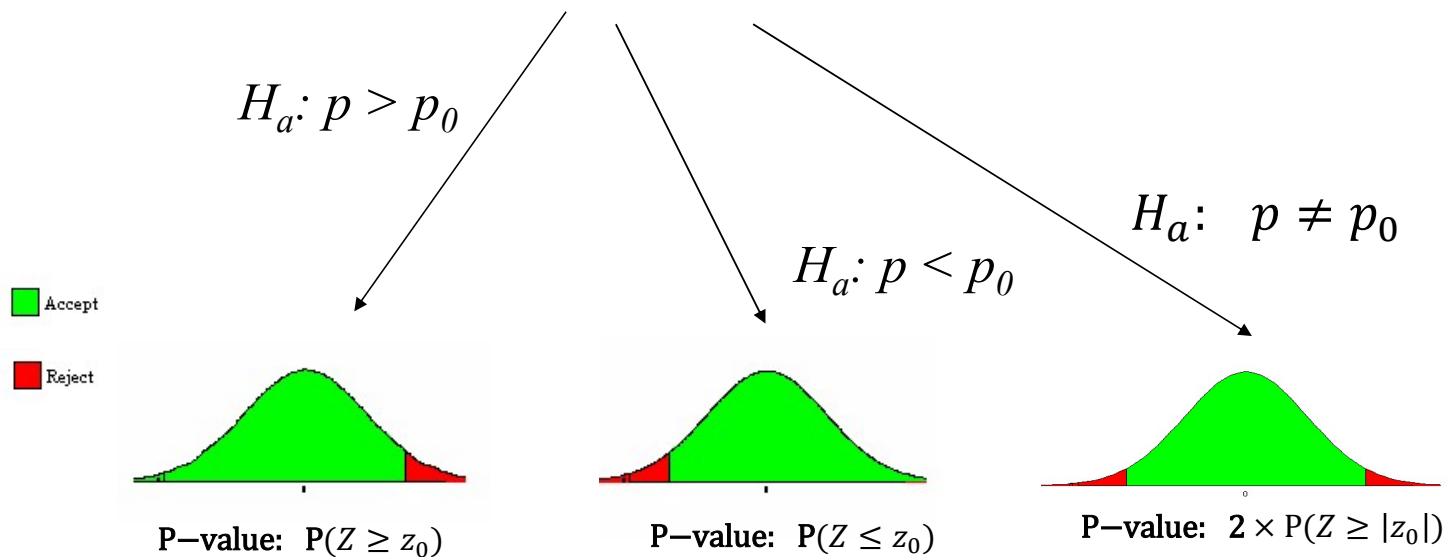


Test statistics, H_a & RR for population proportion p

$$H_0: p = p_0$$

$$\text{Test statistics: } z_0 = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

Test statistics is given
under the assumption
that H_0 is true.



Tests about means of two populations with large sample size

$$H_0: \mu_1 - \mu_2 = \Delta_0$$

Test statistics:

$$Z = \frac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}}}$$

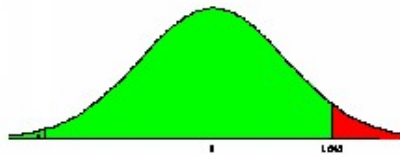
$$H_a: \mu_1 - \mu_2 > \Delta_0$$

$$H_a: \mu_1 - \mu_2 \neq \Delta_0$$

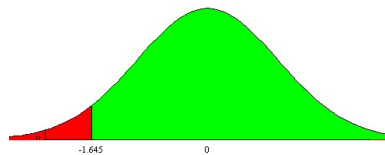
$$H_a: \mu_1 - \mu_2 < \Delta_0$$

Accept

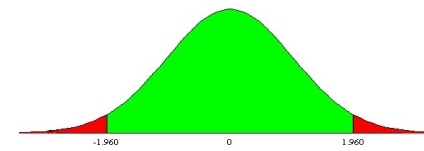
Reject



$$\text{R.R.: } z \geq z_\alpha$$



$$\text{R.R.: } z \leq -z_\alpha$$



$$\text{R.R.: } z \leq -z_{\alpha/2} \\ \text{or } z \geq z_{\alpha/2}$$

Tests about means of two normal populations with unknown variance

$$H_0: \mu_1 - \mu_2 = \Delta_0$$

$$\text{Test statistics: } t = \frac{\bar{x} - \bar{y} - \Delta_0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$$

$$H_a: \mu_1 - \mu_2 > \Delta_0$$

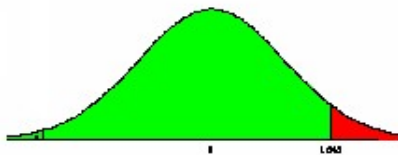
$$H_a: \mu_1 - \mu_2 \neq \Delta_0$$

$$H_a: \mu_1 - \mu_2 < \Delta_0$$

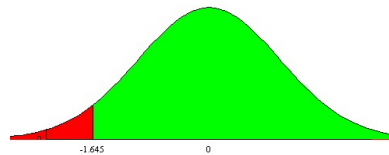
$$v = \left\lceil \frac{\left(\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y} \right)^2}{\frac{\left(\frac{s_X^2}{n_X} \right)^2}{n_X - 1} + \frac{\left(\frac{s_Y^2}{n_Y} \right)^2}{n_Y - 1}} \right\rceil$$

Accept

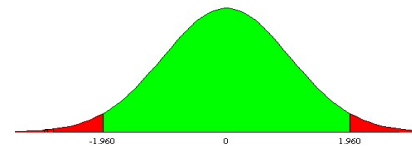
Reject



$$\text{R.R.: } t \geq t_{\alpha, v}$$



$$\text{R.R.: } t \leq -t_{\alpha, v}$$



$$\begin{aligned} \text{R.R.: } & t \leq -t_{\alpha/2, v} \\ \text{or } & t \geq t_{\alpha/2, v} \end{aligned}$$

Tests concerning a difference between population proportions

$$H_0: p_1 - p_2 = \Delta_0$$

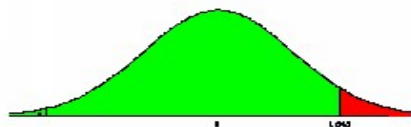
$$\text{Test statistics: } z_0 = \frac{\hat{p}_1 - \hat{p}_2 - \Delta_0}{\sqrt{\frac{\hat{p}_1 \hat{q}_1}{m} + \frac{\hat{p}_2 \hat{q}_2}{n}}}$$

$$H_a: p_1 - p_2 > \Delta_0$$

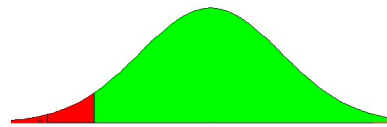
$$H_a: p_1 - p_2 \neq \Delta_0$$

Accept

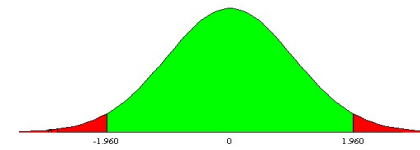
Reject



P-value: $P(Z \geq z_0)$



P-value: $P(Z \leq z_0)$



P-value: $2 \times P(Z \geq |z_0|)$

Most Common $H_0: p_1=p_2$

- Estimators for proportions: $\hat{p}_1 = \frac{X}{m}$ and $\hat{p}_2 = \frac{Y}{n}$
- If H_0 is true, $p = p_1 = p_2$, then p can be estimated by $\hat{p}$.

$$\hat{p} = \frac{m}{m+n} \hat{p}_1 + \frac{n}{m+n} \hat{p}_2 = \frac{X+Y}{m+n}$$

- Then, we change our estimator for the standard deviation:

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{m} + \frac{1}{n}\right)}} \sim N(0,1)$$

Tests for categorical data

1. Uniformity
2. Homogeneity
3. Independence

1. $H_0: p_k = p$ for each category k
2. $H_0: p_{ij} = p_j$ for each column j
3. $H_0: p_{ij} = p_i p_j$ for each cell i, j

$O_{..} = n$: sample size
 k : total number of categories in 1.
 I : total number of rows in 2&3.
 J : total number of columns in 2&3.

1. O_i : # of observed values in category i
2. O_{ij} : # of observed values in cell (i, j)
3. O_{ij} : # of observed values in cell (i, j)

When H_0 is true:

1. E_i : expected # of values in category $i \rightarrow E_i = np_i$
2. E_{ij} : expected # of values in cell $(i, j) \rightarrow E_{ij} = \frac{O_{i.} O_{.j}}{O_{..}}$
3. E_{ij} : expected # of values in cell $(i, j) \rightarrow E_{ij} = \frac{O_{i.} O_{.j}}{O_{..}}$

Assumptions:

1. $E_i \geq 5$ for each category i
2. $E_{ij} \geq 5$ for each cell i, j
3. $E_{ij} \geq 5$ for each cell i, j

If assumptions are not satisfied,
merge columns or rows.

Degrees of freedom when H_0 is true:

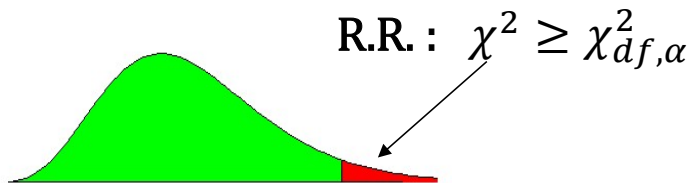
1. $df = k - 1$
2. $df = (I - 1)(J - 1)$
3. $df = (I - 1)(J - 1)$

Tests for categorical data

- | | |
|-----------------|---|
| 1. Uniformity | 1. $H_0: p_k = p$ for each category k |
| 2. Homogeneity | 2. $H_0: p_{ij} = p_j$ for each column j |
| 3. Independence | 3. $H_0: p_{ij} = p_i p_j$ for each cell i, j |

1. O_i : # of observed values in category i
 2. O_{ij} : # of observed values in cell (i, j)
 3. $O_{.j}$: # of observed values in column j

**Rejection
region:**



When H_0 is true:

1. E_i : expected # of values in category $i \rightarrow E_i = np_i$
2. E_{ij} : expected # of values in cell $(i, j) \rightarrow E_{ij} = \frac{O_{i.} O_{.j}}{O_{..}}$
3. E_{ij} : expected # of values in cell $(i, j) \rightarrow E_{ij} = \frac{O_{i.} O_{.j}}{O_{..}}$

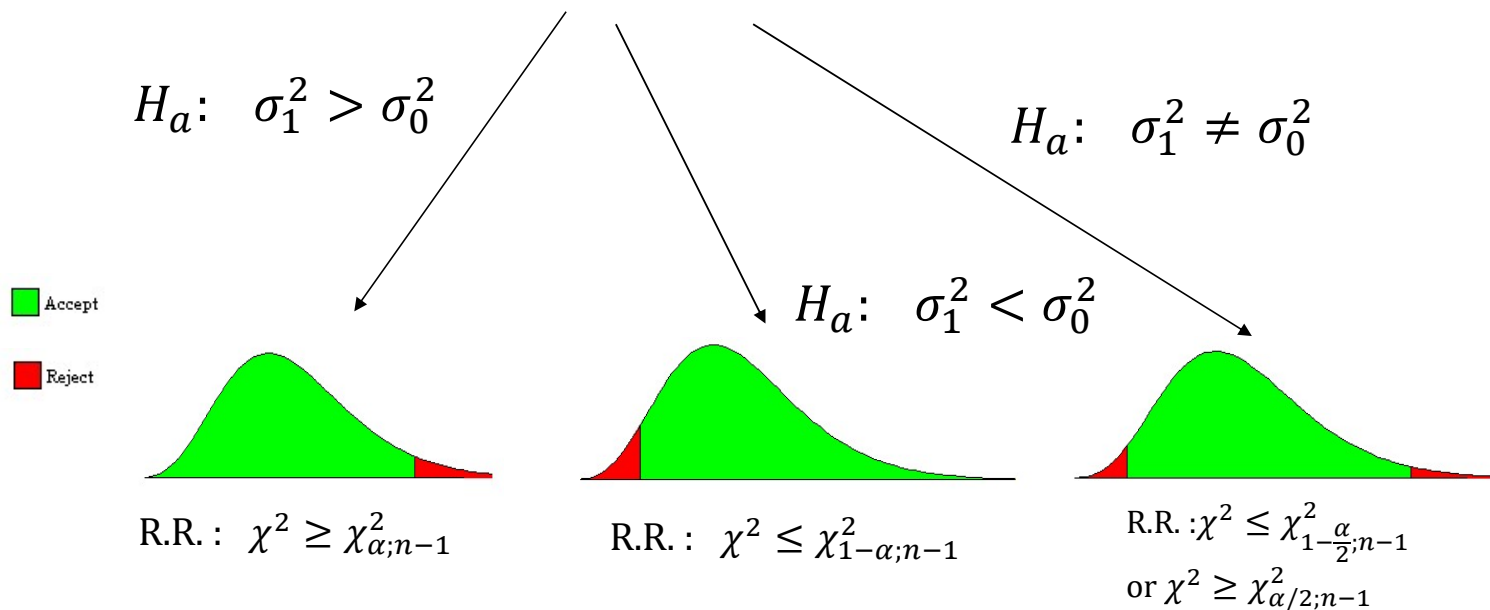
Test statistics and their distributions when H_0 is true:

1. $\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi_{(k-1)}^2$
2. $\chi^2 = \sum_{i=1}^I \sum_{j=1}^J \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi_{(I-1)(J-1)}^2$
3. $\chi^2 = \sum_{i=1}^I \sum_{j=1}^J \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi_{(I-1)(J-1)}^2$

Tests about a population Variance and Standard Deviation of a Normal r.v.

$$H_0: \sigma_1^2 = \sigma_0^2$$

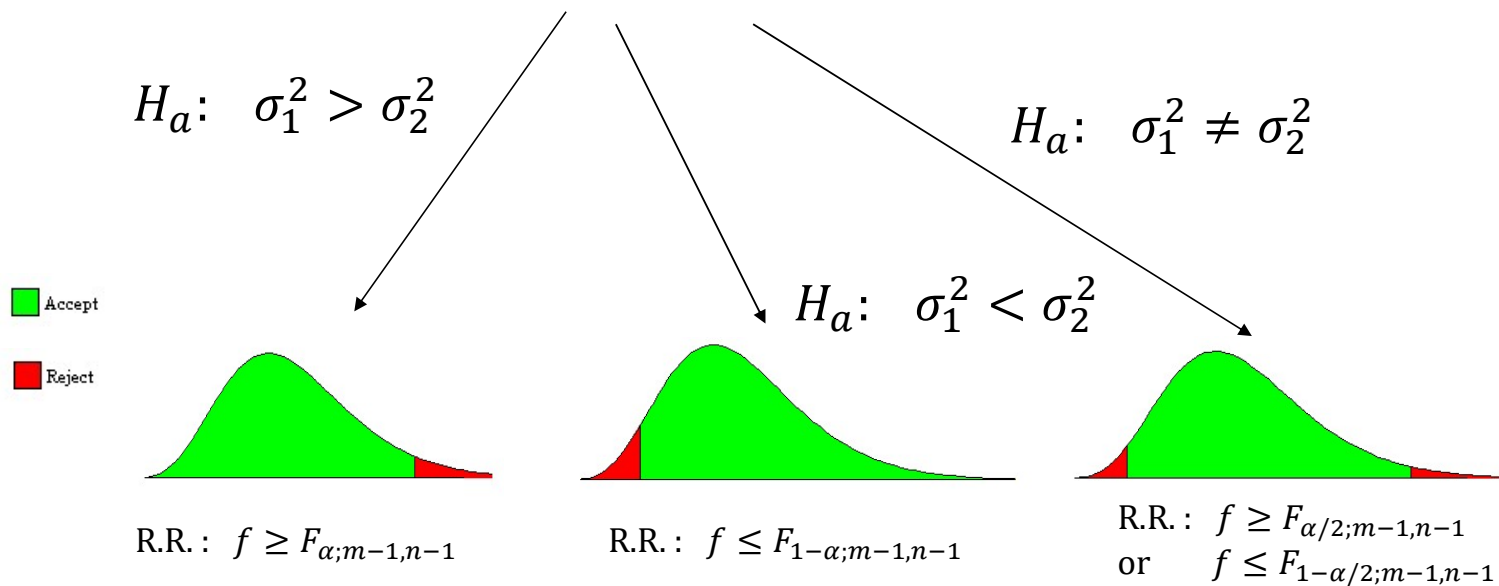
$$\text{Test statistics: } \chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$$



Tests about the equality of two population variances (or standard deviations) of two Normal random variables

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$\text{Test statistics: } F = \frac{S_1^2}{S_2^2}$$



Definition: F distribution

- Let W and Y be χ^2 random variables with u and v degrees of freedom, respectively, then:

$$\frac{W/u}{Y/v} \sim F_{u,v}$$

- Properties:

$$F_{u,v} = \frac{1}{F_{v,u}}$$

$$F_{1-\alpha;u,v} = \frac{1}{F_{\alpha;v,u}}$$

Simple linear regression

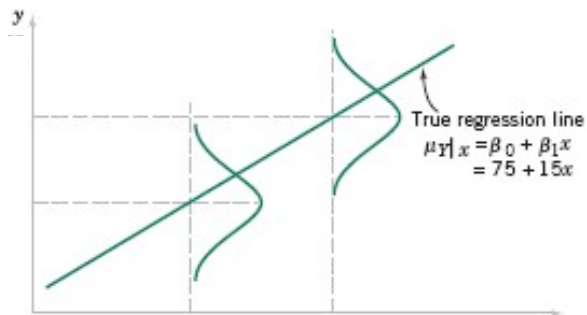
- The sample: $\{(x_i, y_i)\}_{i=1}^n$

independent
variable

dependent
variable

$$\varepsilon \sim N(0, \sigma^2)$$

random
deviation or
random error



True regression line:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

Fitted regression line:

$$y_i = b_0 + b_1 x_i + e_i$$

Least-squares equation or
fitted regression equation.

$$y_i = \underbrace{b_0 + b_1 x_i}_{\hat{y}_i} + \underbrace{e_i}_{\text{residual}}$$

LR Expressions

$$b_0 = \bar{y} - b_1 \bar{x} \sim N \left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) \right)$$

$$b_1 = \frac{S_{xy}}{S_{xx}} \sim N \left(\beta_1, \frac{\sigma^2}{S_{xx}} \right)$$

$$SST = \sum (y_i - \bar{y})^2 = \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{SSR} + \underbrace{\sum (y_i - \hat{y}_i)^2}_{SSE}$$

$$r^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

Correlation Expressions

$$\text{Corr}(X, Y) = \rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

population correlation coefficient

$$r_{xy} = r = \frac{S_{xy}}{\sqrt{S_{xx}} \sqrt{S_{yy}}}$$

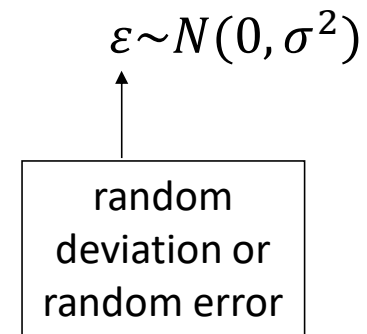
sample correlation coefficient

Multiple linear regression with k independent variables. $k > 1$

- The sample: $\{(x_{1i}, x_{2i}, \dots, x_{ki}, y_i)\}_{i=1}^n$

independent
variables

dependent
variable



True regression line:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} \dots + \beta_k x_{ki} + \varepsilon_i$$

Fitted regression line:

$$y_i = b_0 + b_1 x_{1i} + b_2 x_{2i} \dots + b_k x_{ki} + e_i$$

$$y_i = \overbrace{b_0 + b_1 x_{1i} + b_2 x_{2i} \dots + b_k x_{ki}}^{\hat{y}_i} + \underbrace{e_i}_{\text{residual}}$$

**Least-squares
equation or
fitted regression
equation.**

Multiple regression with k independent variables. $k > 1$:

<i>Source</i>	<i>df</i>	<i>Sum Sq.</i>	<i>Mean Sum Sq.</i>	<i>F</i>
<i>Regression</i>	k	SSR	$MSR = \frac{SSR}{k}$	$\frac{MSR}{MSE}$
<i>Error</i>	$n-(k+1)$	SSE	$MSE = S^2 = \frac{SSE}{n-(k+1)}$	
<i>Total</i>	$n-1$	SST		

$H_0: \beta_i = 0$ for all i
 $H_a: \beta_i \neq 0$ for at least one i

$\xrightarrow{\text{can be tested via:}}$

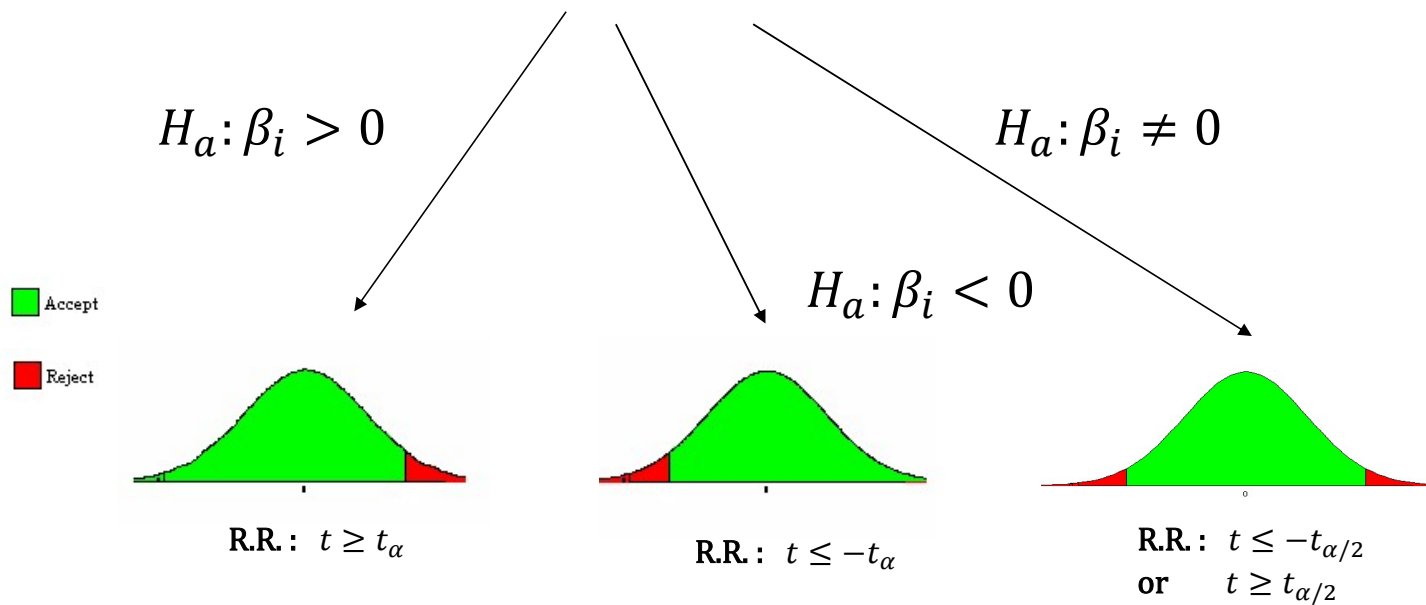
reject H_0 if $F > F_{\alpha, k, n-(k+1)}$.

Overall significance test

Individual Significance Test

Hypothesis testing on β_i

$$H_0: \beta_i = 0 \quad \text{Test statistics: } T = \frac{b_i}{s_{b_i}} \sim t_{n-(k+1)}$$



Summary of Multiple LR (MLR)

The sample:

$$\{(x_{1i}, x_{2i}, \dots, x_{ki}, y_i)\}_{i=1}^n$$

$$Y = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k}_{\text{True regression line}} + \varepsilon \quad \varepsilon \sim N(0, \sigma^2)$$

- Significance of MLR model $H_0: \beta_i = 0$ for all i
 - F-test for the whole model $H_a: \beta_i \neq 0$ for at least one i
 - t-tests for individual independent variables.
 $H_0: \beta_i = 0$
 $H_a: \beta_i \neq 0$
- Percentage of variability explained by the LR model
 - R^2 : no reference to df $\rightarrow$ suitable for simple linear regression (LR) but not MLR
 - Adjusted R^2 : adjusted according to df so as to minimize MSE $\rightarrow$ used for MLR with the same interpretation of R^2 in LR

Evaluation criteria for “best”

$$R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST} \quad \text{coefficient of multiple determination, higher } r^2$$

$$Adj R^2 = R_a^2 = 1 - \frac{SSE/(n-k-1)}{SST/(n-1)} \quad \text{adjusted coefficient of multiple determination, higher } R_a^2$$

Equivalent to minimizing $S^2 = \text{MSE}$.

p : # of all parameters

$$C_p = \frac{(n - p - 1)SSE_{\text{reduced}}(k)}{SSE_{\text{full}}(p)} - (n - 2k - 2) \quad \text{Lower Mallows' } C_p$$

k : # of parameters in a subset

Full vs reduced models

Full model: $y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \beta_{k+1} x_{k+1i} + \cdots + \beta_p x_{pi} + \varepsilon_i$

Reduced model: $y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \varepsilon_i$

$$H_0: \beta_{k+1} = \cdots = \beta_p = 0$$

$$\text{Test statistic: } f = \frac{(SSE_{reduced} - SSE_{full})/(p - k)}{SSE_{full}/(n - p - 1)} \sim F_{p-k, n-(p+1)}$$

Reject H_0 at significance level α if $f > F_{p-k, n-(p+1), \alpha}$